

# Scatterplots with ggplot2

Scatter plots allow us to place points that let us see possible correlations between two features of a data set. Let's see how we can create them with ggplot!

We'll use the built-in mtcars dataset:

In [10]:

```
library('ggplot2')  
df <- mtcars
```

In [13]:

```
head(df)
```

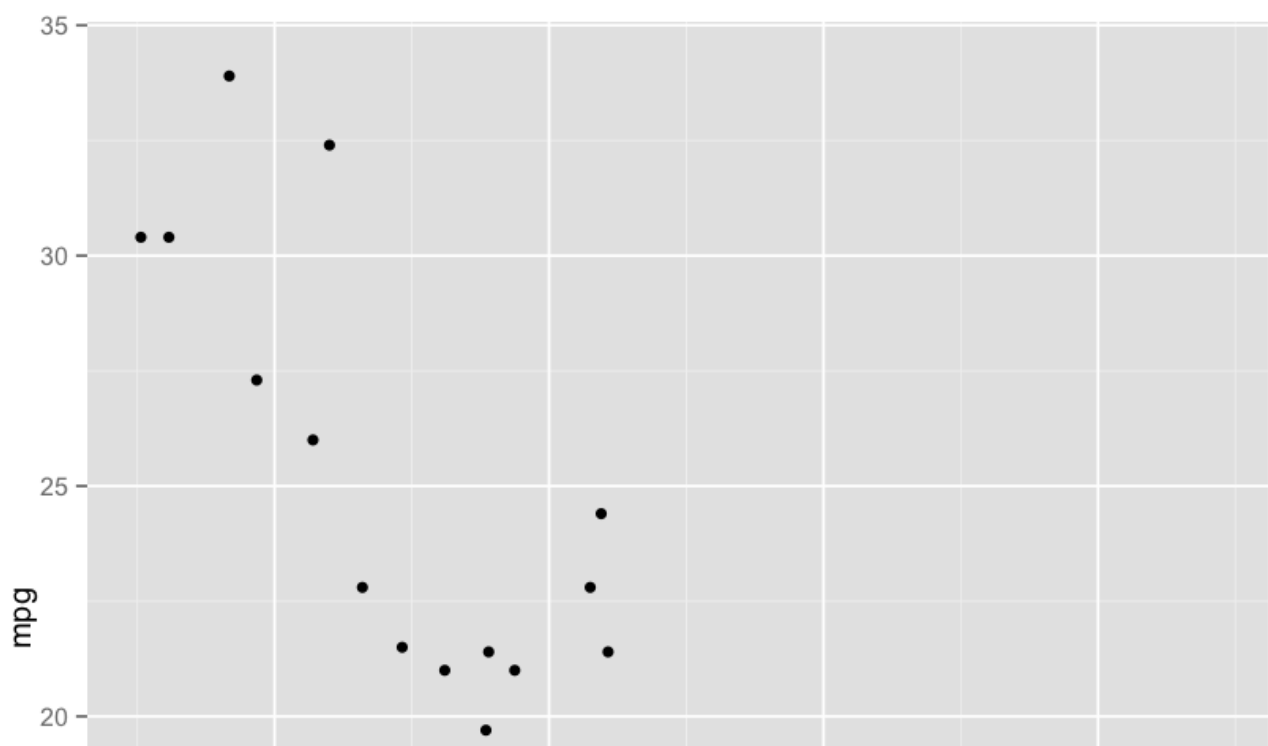
Out[13]:

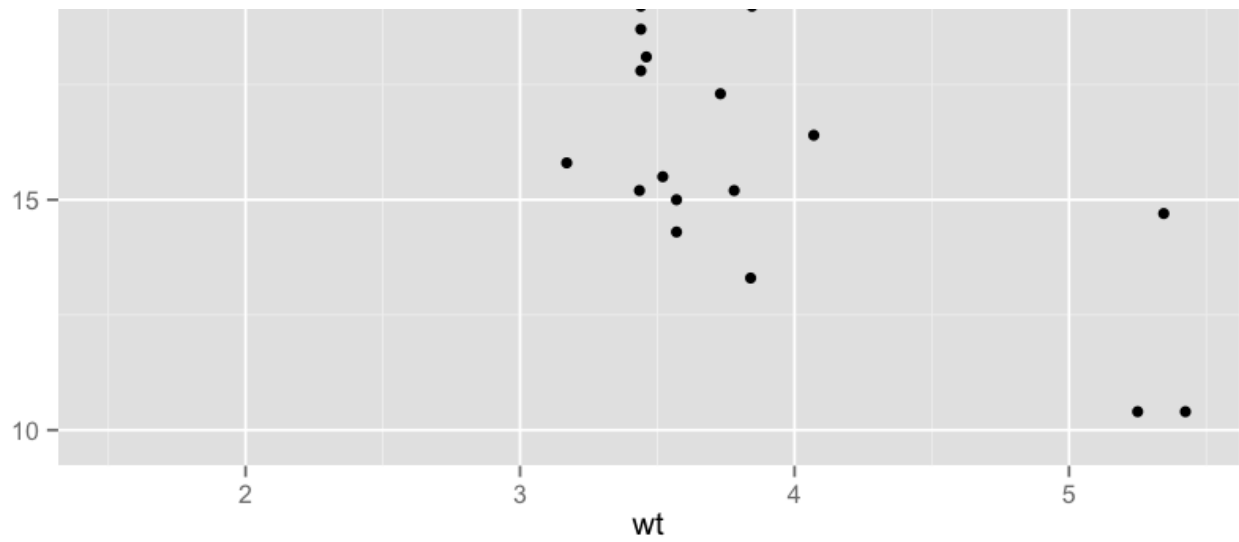
|                          | mpg  | cyl | disp | hp  | drat | wt    | qsec  | vs | am | gear | carb |
|--------------------------|------|-----|------|-----|------|-------|-------|----|----|------|------|
| <b>Mazda RX4</b>         | 21   | 6   | 160  | 110 | 3.9  | 2.62  | 16.46 | 0  | 1  | 4    | 4    |
| <b>Mazda RX4 Wag</b>     | 21   | 6   | 160  | 110 | 3.9  | 2.875 | 17.02 | 0  | 1  | 4    | 4    |
| <b>Datsun 710</b>        | 22.8 | 4   | 108  | 93  | 3.85 | 2.32  | 18.61 | 1  | 1  | 4    | 1    |
| <b>Hornet 4 Drive</b>    | 21.4 | 6   | 258  | 110 | 3.08 | 3.215 | 19.44 | 1  | 0  | 3    | 1    |
| <b>Hornet Sportabout</b> | 18.7 | 8   | 360  | 175 | 3.15 | 3.44  | 17.02 | 0  | 0  | 3    | 2    |
| <b>Valiant</b>           | 18.1 | 6   | 225  | 105 | 2.76 | 3.46  | 20.22 | 1  | 0  | 3    | 1    |

## qplot()

In [14]:

```
qplot(wt,mpg,data=df)
```



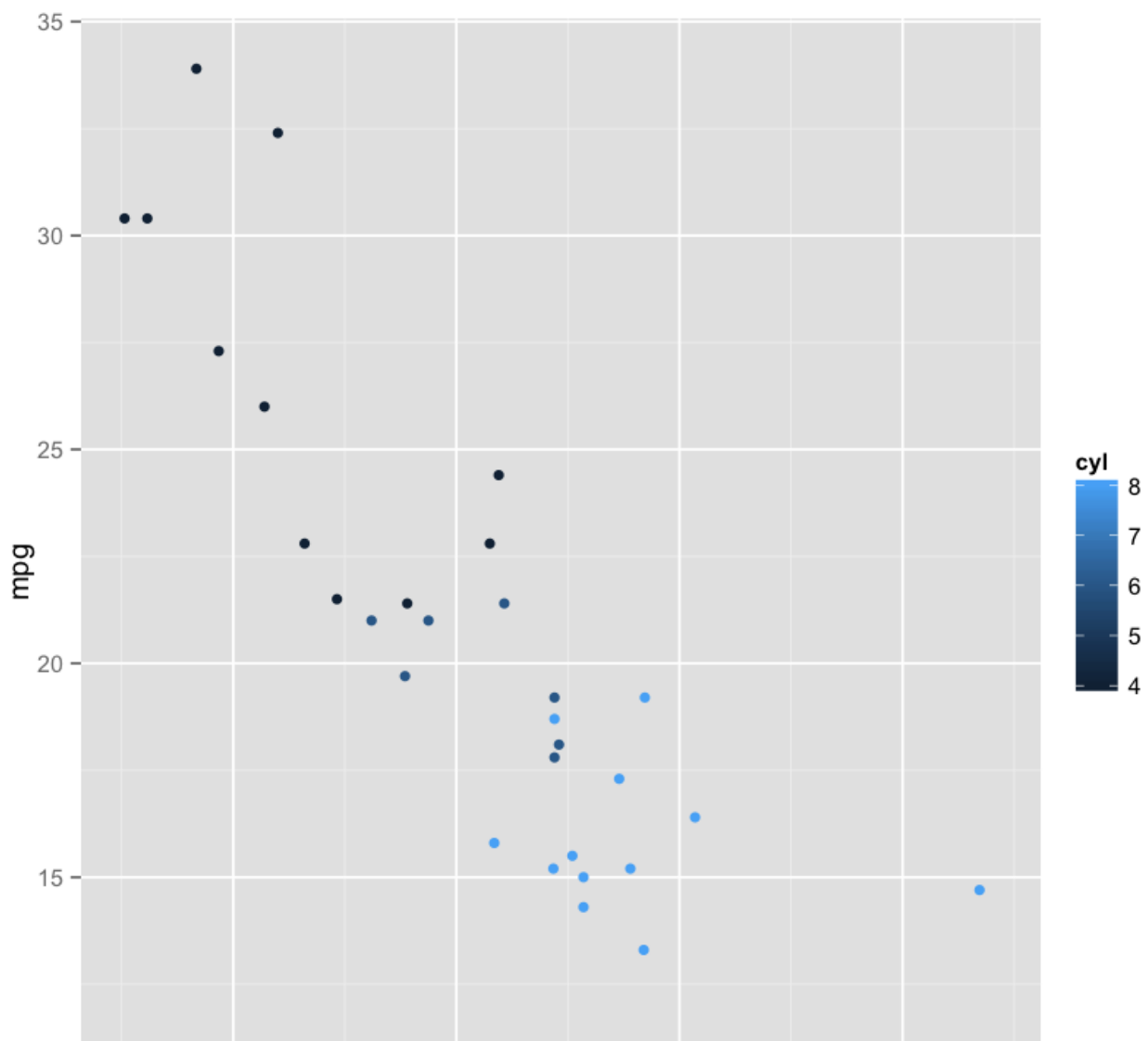


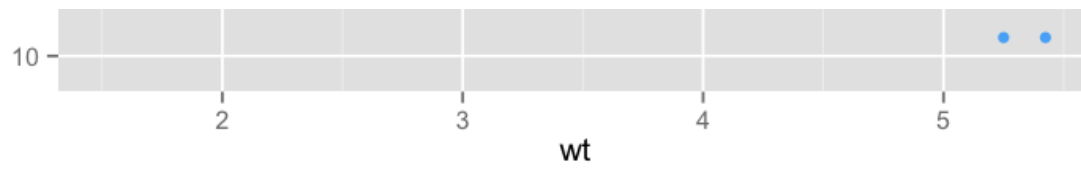
## Adding a 3rd feature

We can add a third feature by adding a color gradient on each point, or by resizing each point based on their value of this 3rd feature. For example:

In [15]:

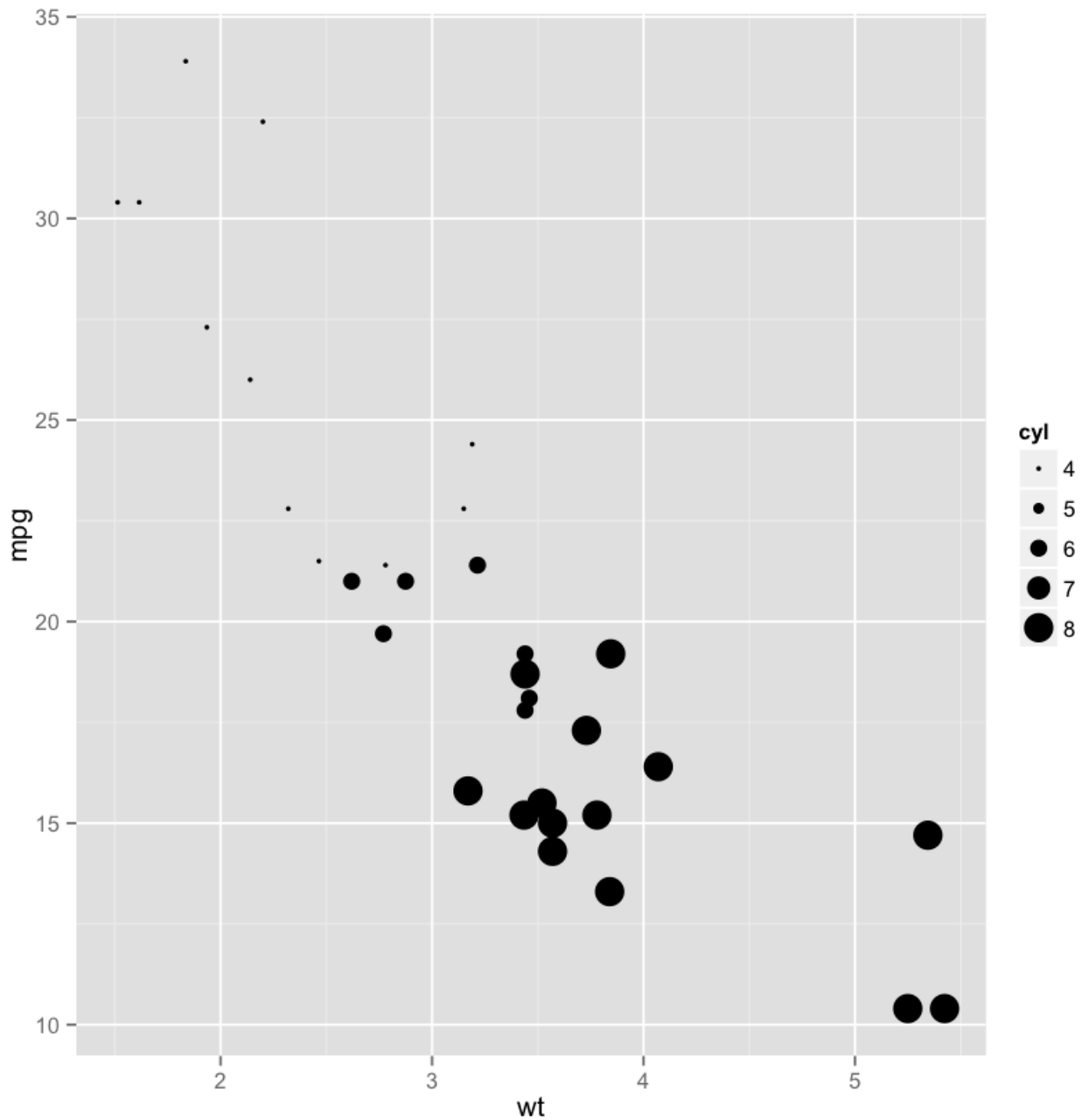
```
qplot(wt,mpg,data=df,color=cyl)
```





In [17]:

```
qplot(wt,mpg,data=df,size=cyl)
```



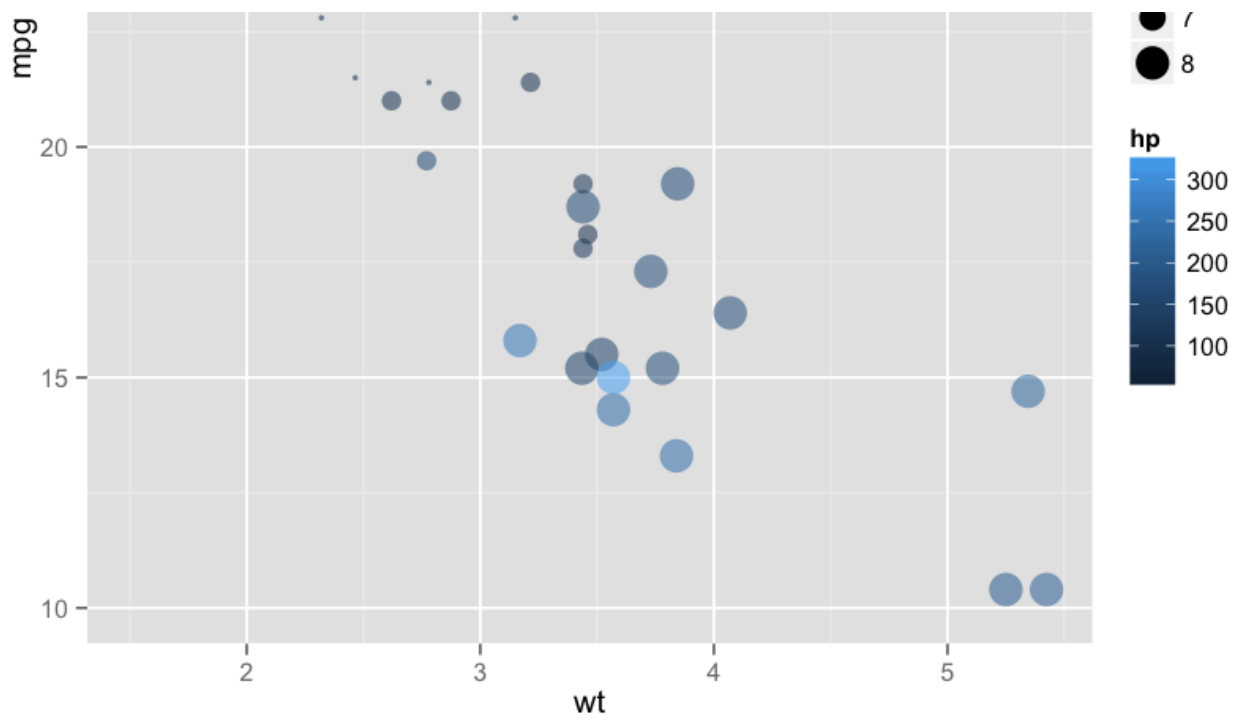
## Or both

In [18]:

```
qplot(wt,mpg,data=df,size=cyl,color=cyl)
```





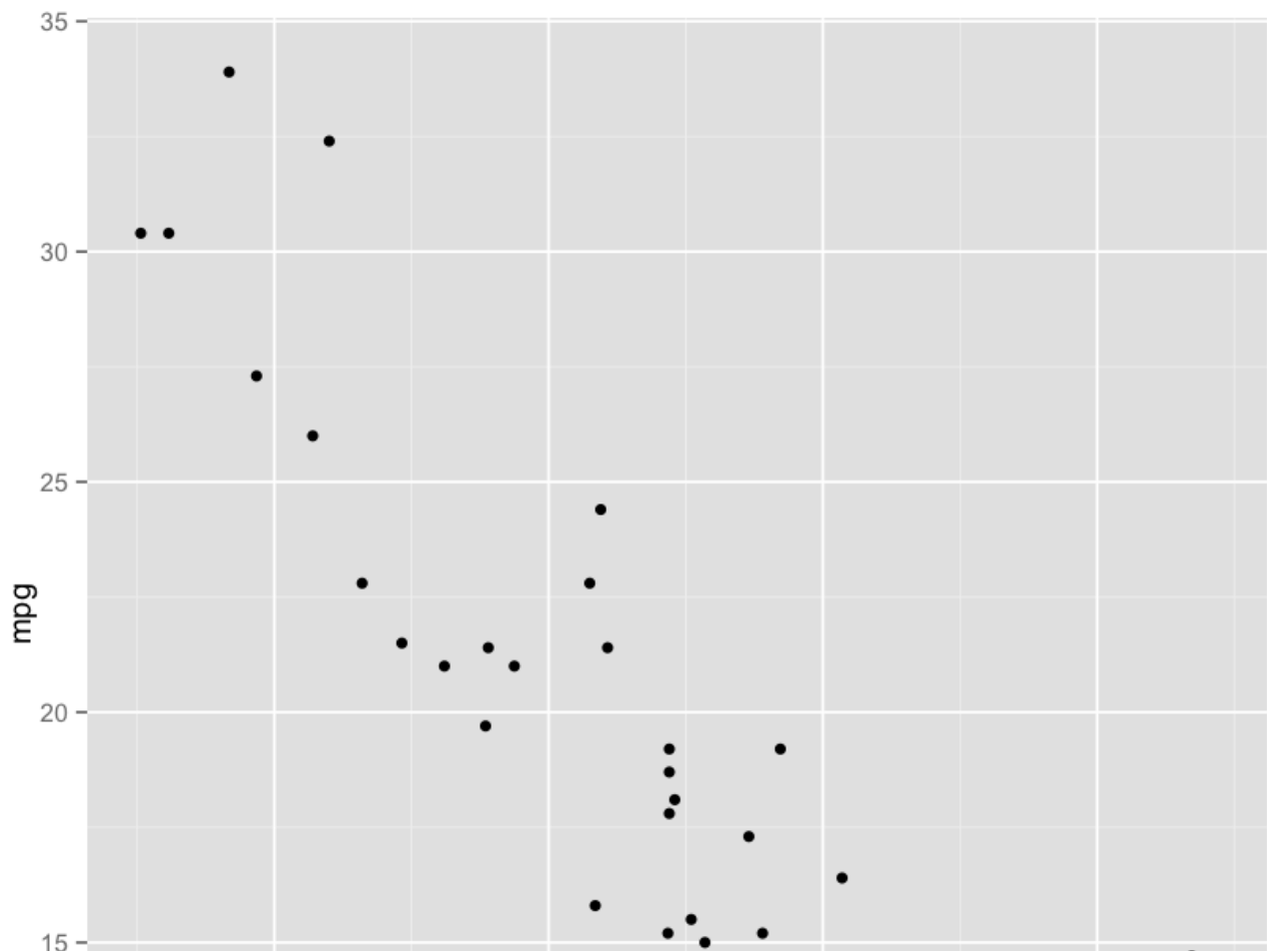


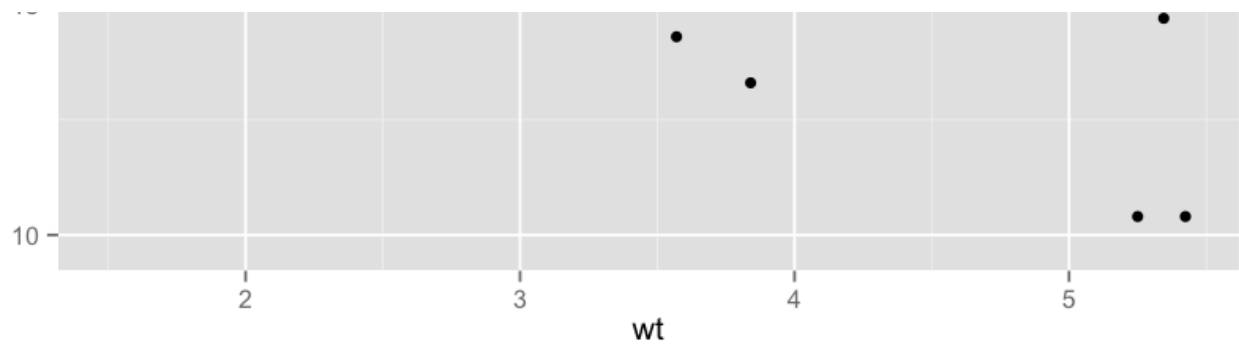
## ggplot()

Now let's see how to get more control by using ggplot():

In [30]:

```
p1 <- ggplot(data=df, aes(x = wt, y=mpg))
p1 + geom_point()
```

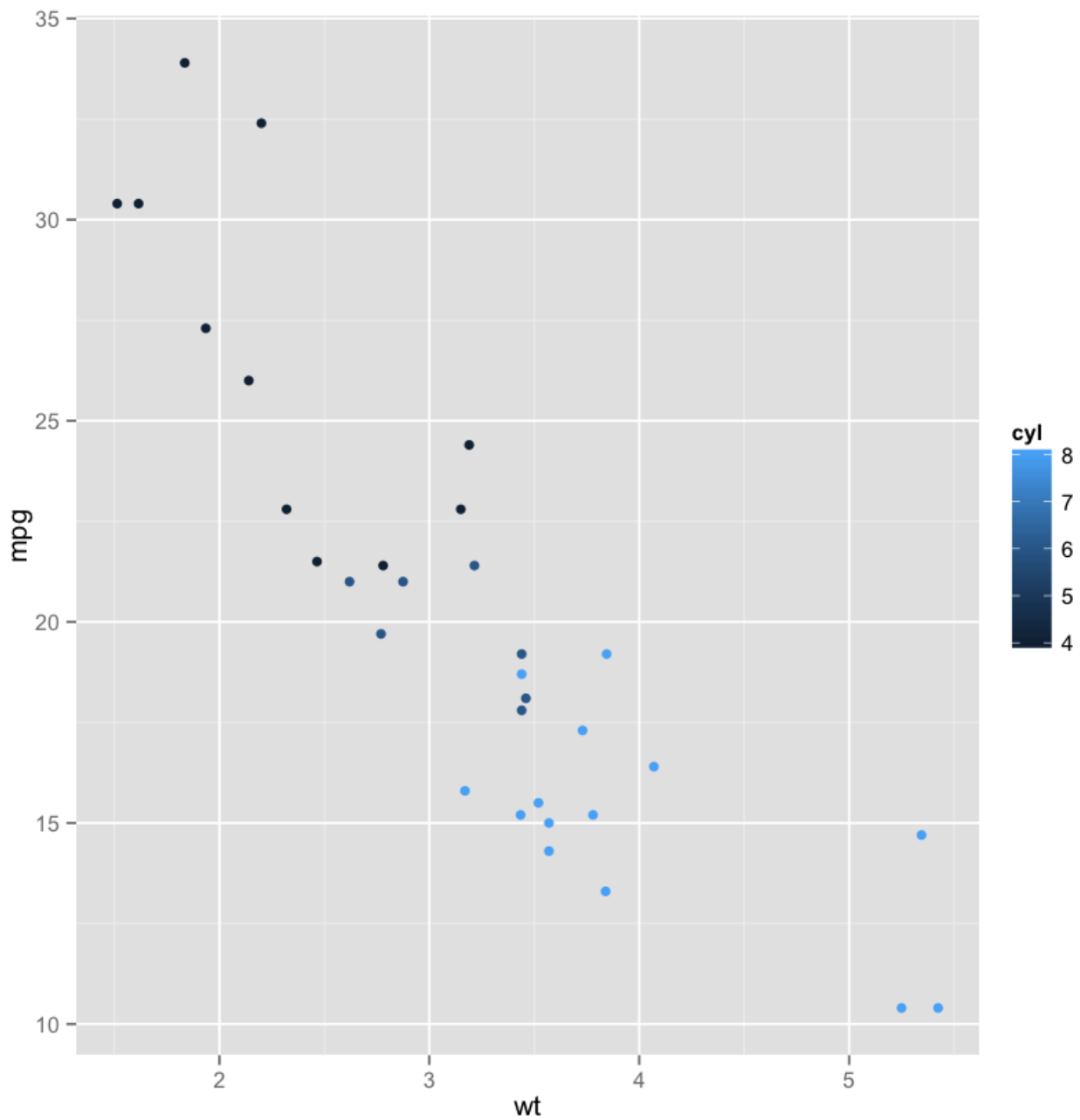




## Adding 3rd feature

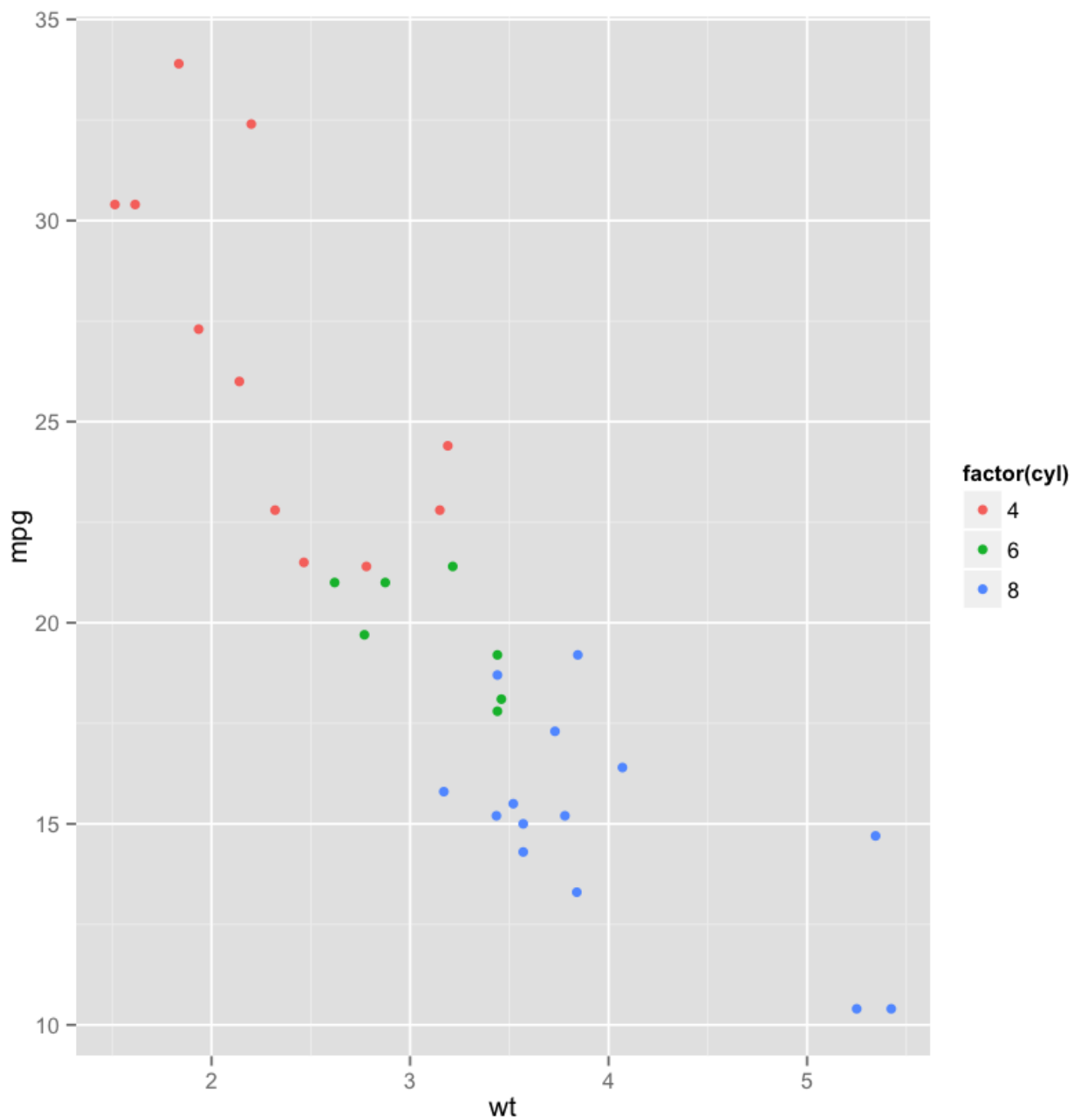
In [32]:

```
p1 <- ggplot(data=df, aes(x = wt, y=mpg))  
p1 + geom_point(aes(color=cyl))
```



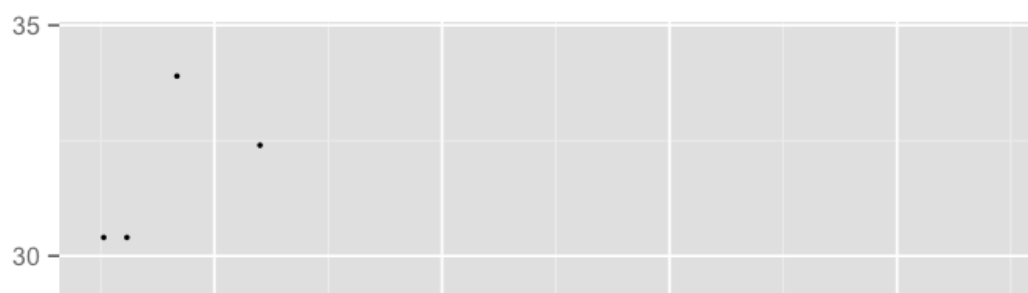
In [37]:

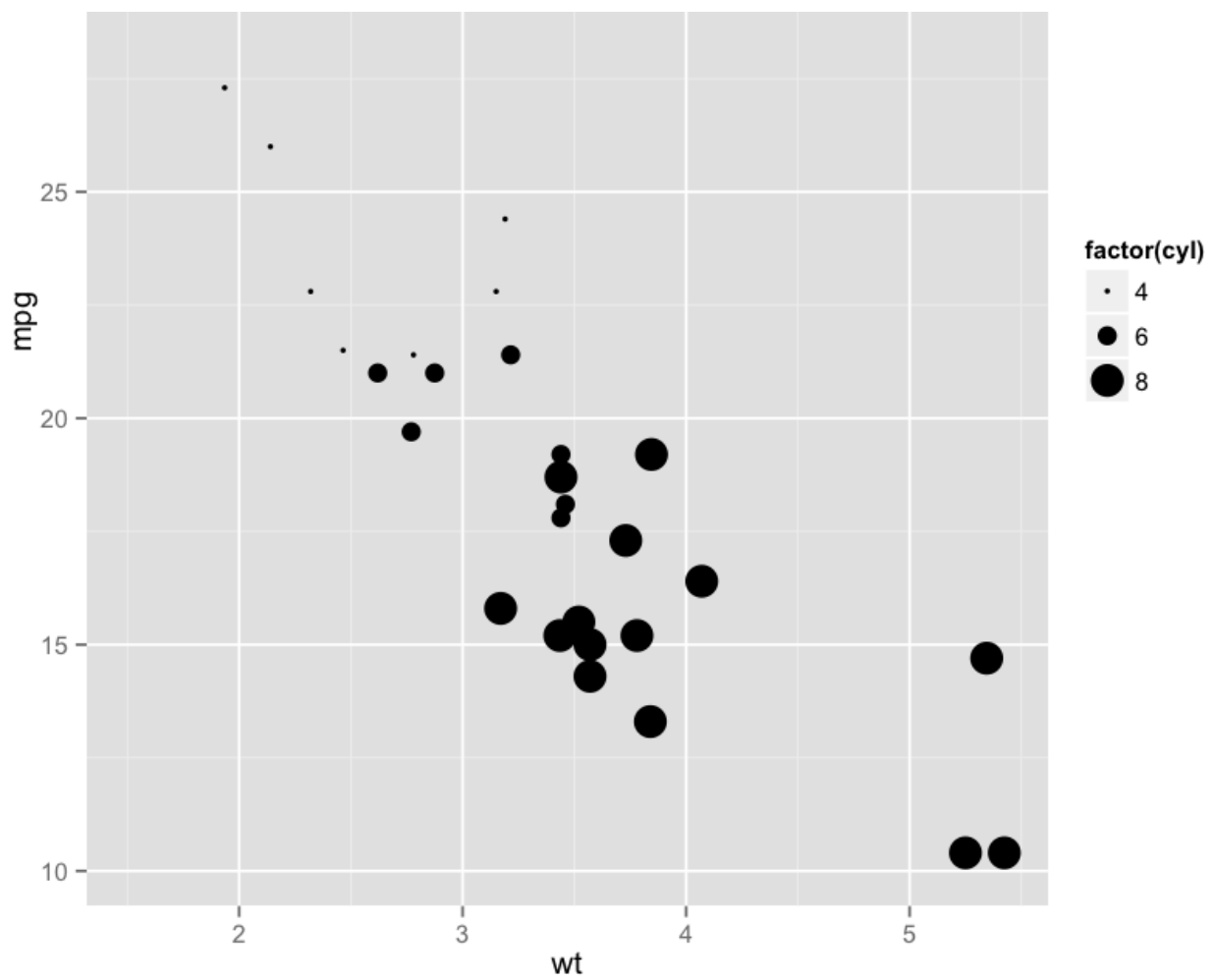
```
p1 <- ggplot(data=df,aes(x = wt,y=mpg))  
p1 + geom_point(aes(color=factor(cyl)))
```



In [38]:

```
p1 <- ggplot(data=df,aes(x = wt,y=mpg))  
p1 + geom_point(aes(size=factor(cyl)))
```

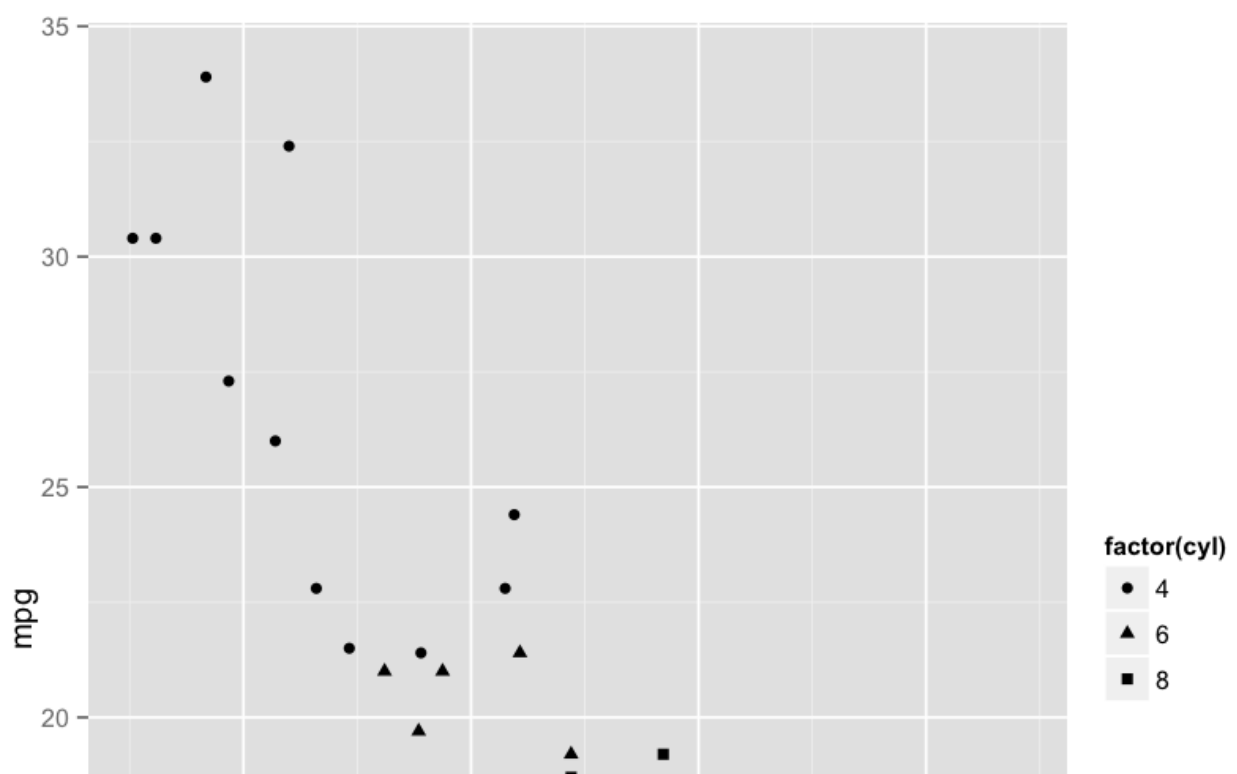




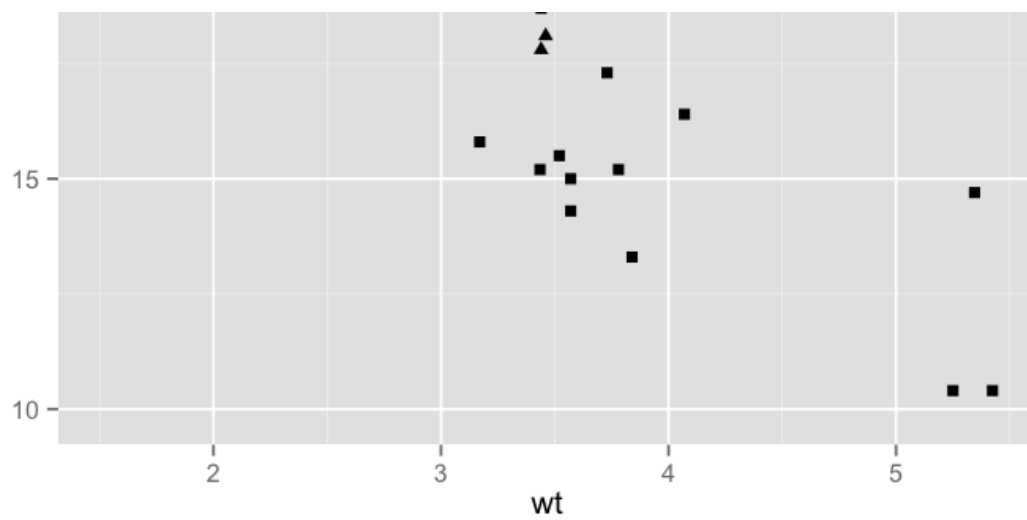
In [47]:

```
# With Shapes
pl <- ggplot(data=df,aes(x = wt,y=mpg))

pl + geom_point(aes(shape=factor(cyl)))
```



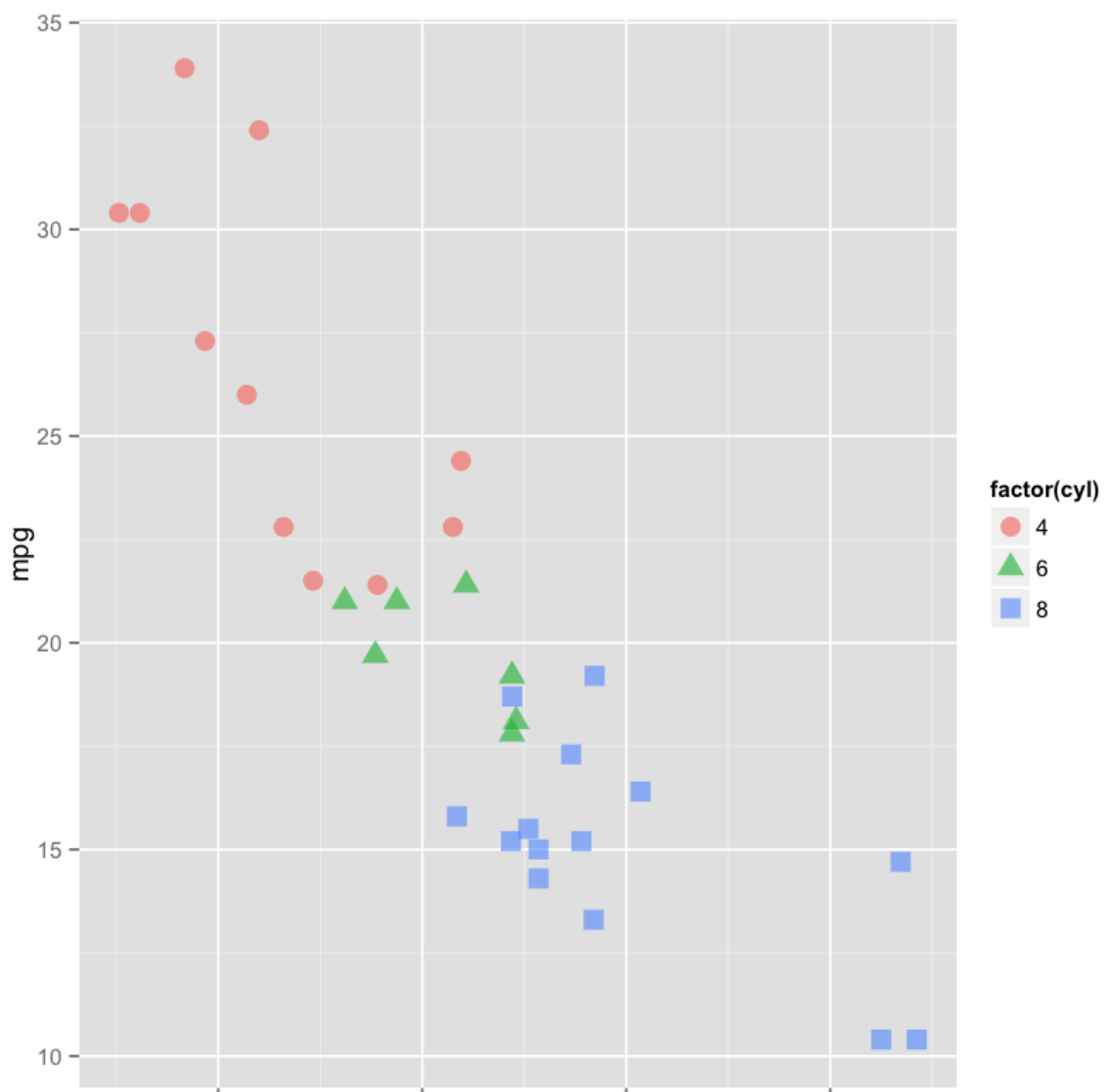




In [49]:

```
# Better version
# With Shapes
pl <- ggplot(data=df,aes(x = wt,y=mpg))

pl + geom_point(aes(shape=factor(cyl),color=factor(cyl)),size=4,alpha=0.6)
```



2

3

wt

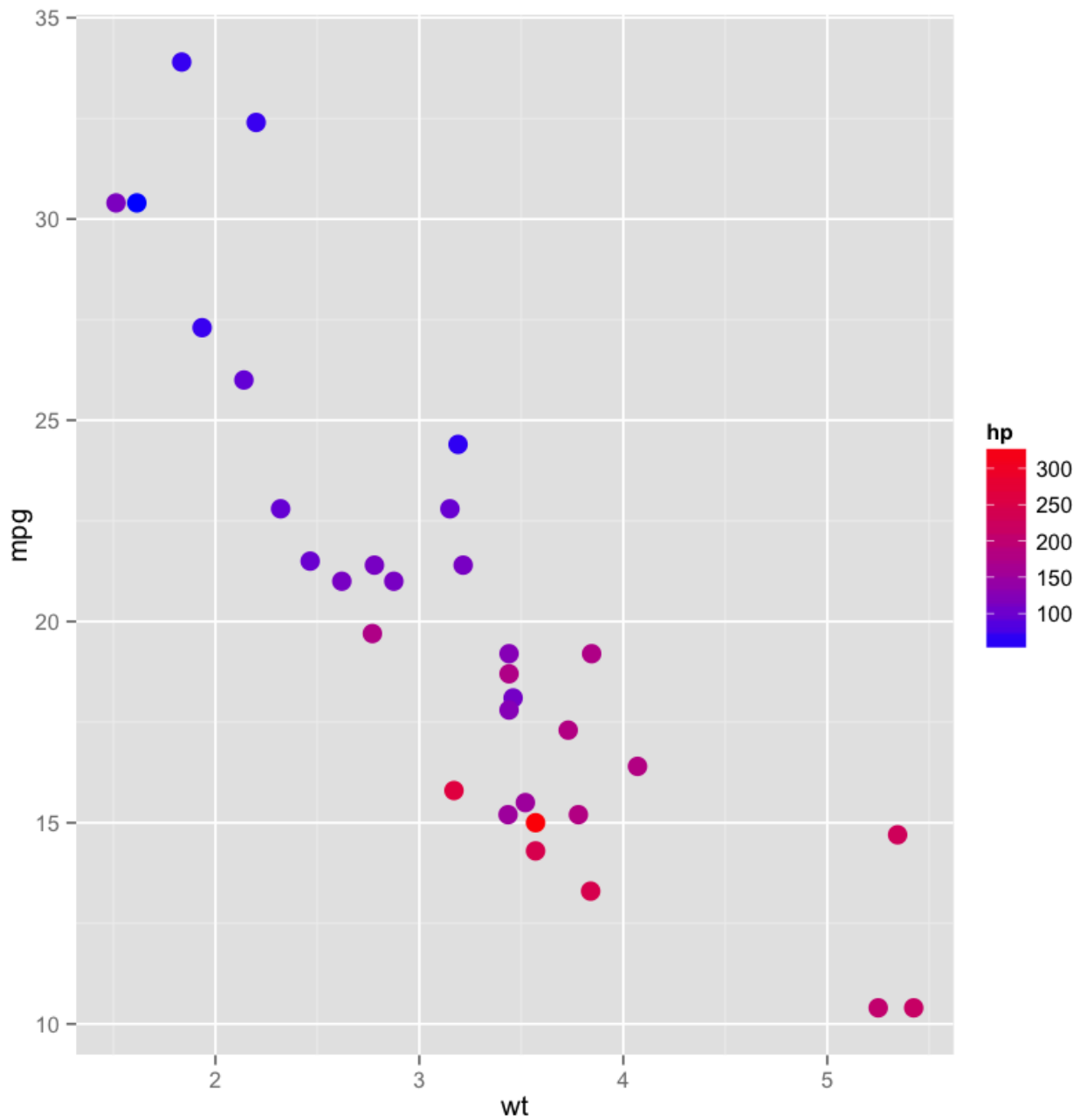
4

5

## Gradient Scales

In [50]:

```
pl + geom_point(aes(colour = hp), size=4) + scale_colour_gradient(high='red', low = "blue")
```



Great! That's it for scatterplots, remember to reference the cheat sheet or the documentation for more details!